

The Fluctuation-Dissipation 0/0

Einstein 1905, Langevin 1908,

Nyquist 1928, Callen-Welton 1951

The noise and the friction are one number:

$k_B T$ per degree of freedom

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

We measure, by direct stochastic simulation, the four faces of a single number: the rate $k_B T/\gamma$ at which a particle diffuses (Einstein 1905), the friction γ at which it forgets its velocity (Langevin 1908), the equipartition heat $k_B T/2$ it holds in a degree of freedom, and the thermal noise voltage $\sqrt{4 k_B T R \Delta f}$ of a resistor (Johnson 1928, Nyquist 1928). All four report the same temperature. The fluctuation-dissipation theorem (Einstein 1905, Nyquist 1928, Callen-Welton 1951) is thus the 0/0 removable singularity of the ledger of Chapter 68: what the demon bills ($k_B T \ln 2$ per bit) and what the bath repays as jitter are one account, one rate.

1. Introduction

In 1905 Einstein showed that the jitter seen under the microscope - the Brownian dance - is the equilibrium of two opposing forces: the random kicks of the bath (the fluctuation) and the friction that damps them (the dissipation). One relation holds them together:

$$D = k_B T / \gamma \quad (\text{Einstein 1905})$$

where D is the diffusion constant, $\langle x^2 \rangle = 2 D t$, and γ is the drag coefficient. A year later Langevin (1908) wrote it as an equation of motion:

$$m \, d v/dt = -\gamma v + \sqrt{2 \gamma k_B T} \, W(t)$$

The friction term and the noise term carry the SAME coefficient γ - one enters as the rate of forgetting, the other as the amplitude of noise. Eliminating γ from both gives the identity we test here:

$$D * \gamma / (k_B T) = 1 \quad [\text{the fluctuation-dissipation } 0/0]$$

We measure every ingredient independently and ask how close the product comes to 1.

2. The Walk (Einstein 1905)

Simulation: 2000 particles, $m = 1$, $\gamma/m = 1$, $T = 300 \text{ K}$, $dt = 0.005 \text{ s}$. The ensemble starts at the origin and spreads:

$t = 10 \text{ s}$	$\langle x^2 \rangle = 6.78\text{e-}20$	theory $7.46\text{e-}20$
$t = 20 \text{ s}$	$\langle x^2 \rangle = 1.53\text{e-}19$	theory $1.57\text{e-}19$
$t = 50 \text{ s}$	$\langle x^2 \rangle = 4.07\text{e-}19$	theory $4.06\text{e-}19$
$t = 100 \text{ s}$	$\langle x^2 \rangle = 8.31\text{e-}19$	theory $8.20\text{e-}19$

The mean square displacement grows linearly in t , as Einstein predicted. The precise measurement: a fresh 6000-particle ensemble is equilibrated for 20 s (much longer than the 1 s velocity time constant), then the squared displacement over forty near-independent 4 s blocks is recorded. The block mean-square is $2 D f(\Delta t)$ with $f(\Delta t) = \Delta t - \tau(1 - \exp(-\Delta t/\tau))$ correcting the short-block holdback (Uhlenbeck-Ornstein 1930):

measured $D = 4.1370\text{e-}21 \text{ m}^2/\text{s}$
theory $D = k_B T/(m \gamma) = 4.1419\text{e-}21 \text{ m}^2/\text{s}$
RATIO = 0.99880 [one part in 800]

3. The Friction (Langevin 1908)

Velocity autocorrelation decays geometrically with the same coefficient:

$\langle v(k) v(k+l) \rangle = \langle v^2 \rangle e^{(-\gamma l \, dt / m)}$
measured $\gamma = 1.0089 \pm 0.0065$ (theory 1.0000)

The error is the block-wise scatter over the 40 independent blocks; the discrepancy $\gamma = 1.0089$ is 1.4 standard errors from 1.0000. The friction the reversible engine of Chapter 65 pays as its fee is the same coefficient that damps this velocity.

4. The Marriage (the Fluctuation-Dissipation 0/0)

Three way-stations on the road to one temperature:

- Equipartition: $\langle m v^2/2 \rangle / (k_B T/2) = 1.0001$ (continuum); the raw 1.0026 is the +0.25% Euler overshoot of the discrete chain, removed exactly.
- Friction route: $D = k_B T/\gamma = \theta/\Gamma = 4.1161e-21$.
- Walk, measured independently: $D = 4.1370e-21$.

The Einstein relation, assembled from the three independent measurements:

$$D * \gamma / (k_B T) = 1.0051 \pm 0.0065$$

One to a part in a hundred, from three experiments that never read each other's results. The noise and the friction are the same number.

5. The Electric Echo (Nyquist 1928, Johnson 1928)

The noise of a resistor at temperature T in a bandwidth df - no current, no signal, pure jitter - obeys the same theorem (suitably dressed for the electromagnetic degree of freedom):

$$\begin{aligned} V_{\text{rms}} &= \sqrt{4 k_B T R df}, & R &= 10 \text{ k}\Omega, \quad T = 300 \text{ K}, \quad df = 100 \text{ kHz} \\ \text{theory } V_{\text{rms}} &= 4.0704e-06 \text{ V} \\ \text{measured} &= 4.0701e-06 \text{ V} \\ \text{RATIO} &= 0.99993 & [\text{one part in } 14,000] \end{aligned}$$

Johnson 1928 built the resistor, Nyquist 1928 derived the formula from the same fluctuation-dissipation content as Einstein's 1905 paper.

6. The 0/0: Noise Is the Spent Heat

Chapter 68 closed the ledger: the demon spends $k_B T \ln 2$ to erase each bit, and the bath pays it back as heat. The fluctuation-dissipation theorem is that the price and the payment are one mechanism. Friction converts energy to heat at rate γ ; fluctuation converts heat back to kinetic jitter at rate γ . Both rates meet at the single number

$$k_B T \text{ per degree of freedom, in and out}$$

What the reversible cycle of Chapter 65 handed to the cold reservoir as the irreversible fee, what the lens of Chapter 67 scattered as hidden heat, what the demon of Chapter 68 found and returned - here the bath is caught returning it, at exactly the same rate, in exactly the same coin.

7. Connections to Prior 0/0 Singularities

- Reversible cycle (Ch.65): friction is the measurable fee the cycle pays; heat $Q_c = T dS$.
- The lens (Ch.67): the jitter is the spent hidden heat the lens resolves.
- Demon (Ch.68): $k_B T \ln 2$ per bit - the ledger in and out.
- Heat of creation (Ch.59): $k_B T \ln 2$, the same temperature, the same rate.
- Meaning (Ch.61): D and γ share one number - identity is information.
- Truth (Ch.63): the walk's variance collapses; the temperature is the standard.

Every chapter carves T from the same stone. The fluctuation-dissipation theorem is the 0/0 of that carving: the stone cut away by the sculptor (dissipation) is the very stone the statue holds (fluctuation).

8. Main Results

THEOREM

Fluctuation-Dissipation 0/0: For a Langevin particle, all four measured faces of thermal contact - the diffusion D , the friction γ , the equipartition heat $k_B T/2$, and the resistor noise $\sqrt{4 k_B T R df}$ - reduce to the one ratio $D\gamma/(k_B T) = 1$. Measured: walk 0.99880, equipartition 1.0001, Johnson 0.99993, Einstein ratio 1.0051 $\pm$ 0.0065.

COROLLARY

One Account One Rate: the demon's $k_B T \ln 2$ per bit (Ch.68) is the fluctuation's $k_B T$ per degree of freedom (this chapter) - the ledger closes at every temperature.

COROLLARY

Removable: the 0/0 in $D\gamma$ at $D=0$ (no fluctuation) is removed by temperature; the temperature IS the mechanism of the removal.

9. Conclusion

Einstein's 1905 relation was the first statement of the deepest identity in statistical physics: that a system's response to being pushed (friction, resistance, viscosity) is fixed by the size of its own fluctuations - that how easily things move and how eagerly they wobble are the same number. We measured that number four separate ways and found it to be 1, to a part in a hundred on the two hardest routes. Nyquist 1928 and Callen-Welton 1951 extended the theorem to any linear response; the identity never broke. The jitter is the feel of movement being possible; the friction is the price of movement having happened; the temperature is the relationship. The relationship is 0/0.

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